You don’t have to choose your career today.

But how do you choose a career you’ve never seen?

Careers in
AI & Full Stack Development

Education, working life and career opportunities.

AUDIENCE
Classes IX to XII

VENUE
KV RK Puram, Sector 2

DEPARTMENT
DEPFE, NCERT, New Delhi

Tushar Gaurav

Software Engineer

Orangewood Labs, Noida, India

tushgaurav.com

B.Tech in Computer Science Engineering
Noida Institute of Engineering and Technology, 2020 to 2024

Full Stack

Cloud & DevOps

AI

Scan to visit my website

A few terms you'll hear today

Think of opening your school's website.

Client

The browser or app that asks for information—like the browser on your phone.

Server

A computer or program that receives requests and sends back a response.

Frontend

The part you see and interact with: pages, menus, buttons and forms.

Backend

The logic behind the scenes: checking a login or processing a form.

Database

An organised store of information, such as student records or school notices.

Cloud

Computers and services you use over the internet to run websites and store data.

Full Stack Developer

Builds both the frontend and backend of a website—and connects the pieces into a working application.

Write the code

Build pages, buttons and forms.
Write backend logic for things like logins and requests.

Connect the data

Connect the application to databases.
Save, find and update the information people need.

Test & launch

Test features and debug—find and fix errors.
Deploy websites to the cloud so people can use them.

Work environment and personal traits

Places and hours

- Employment in **IT companies**
- Typically **5 to 6 days a week**
- **8 to 9 hours per day**
- Shift systems may apply

Personality traits

- Good with computers
- Enjoy analysing problems and situations
- Prefer clear instructions to follow
- Comfortable communicating with others

Career options

- Work as an employee
- Pursue freelancing
- Work for your own company
- Opportunities exist for people with disabilities

Fresher Developer → Full Stack Developer → Senior Full Stack Developer

Education and course fees

Educational pathway

- Complete **10+2 in Science: Physics, Chemistry and Mathematics**
- Bachelor's degree: **B.Tech / B.Sc / BCA / B.E.** in Computer Science / Information System
- Or a bachelor's followed by **M.Tech / M.Sc / MCA / M.E.** in the same field

Course fee

₹55,000 – ₹10,00,000

- Indicative course-fee range
- Fees vary by institution; do not assume this is an annual fee
- Check course duration and current admission requirements before enrolling

Individual programmes may have different eligibility rules.

Expected income and funding

Approximate monthly income

₹24,000 – ₹1,67,000

- **Per month**; varies by experience, employer and location
- Indicative and subject to change; not a guaranteed starting salary

Scholarships and loans

- Engineering-specific and institute merit scholarships
- National Scholarship Portal: central, UGC / AICTE and state schemes
- Vidya Lakshmi education-loan portal
- Bank education loans and some state student-credit schemes; check eligibility and terms

Scholarship availability and eligibility can change. Verify current schemes and application deadlines.

Artificial Intelligence Engineer

Uses AI and machine learning to develop applications and systems that help organisations operate more efficiently.

Myths about AI

Myth: AI is always correct

AI can generate convincing but incorrect answers. Testing, reliable sources and human review still matter.

Myth: AI will replace every job

AI can automate tasks, but its impact varies by role. Human judgement, responsibility and domain knowledge remain important.

Myth: AI engineering is just prompting

Building dependable AI systems also requires programming, data handling, evaluation, deployment and monitoring.

Myth: Only an AI degree leads into AI

Several routes can lead into AI, including Computer Science, Mathematics, Statistics, IT and AI / ML study.

Routes into AI engineering

A common starting point is 10+2 in Science: Physics, Chemistry and Mathematics.

Degree or diploma

- Bachelor's in **AI, Mathematics, Statistics, Computer Science or Information Technology**
- Or a **diploma in AI and Machine Learning**
- Another route: a bachelor's in Maths / Statistics / CS / IT, followed by an **AI-related online certificate**

Postgraduate route

- A bachelor's followed by a **master's in a relevant field**
- Examples include CS & Engineering, AI, ML, Robotics, Data Science and Data Engineering
- Other allied fields include Electronics, Signal Processing and Computational Linguistics

Admission may involve national, state or institute entrance exams, including JEE Main / Advanced and WBJEE. Requirements vary by course; verify current eligibility and duration.

NATURE OF WORK

Responsibilities and tools

Practical examples of AI engineering work; responsibilities vary by team and product.

Core Responsibilities

- Design, train and fine tune **machine learning and deep learning** models
- Collect, clean and prepare data, then engineer features
- Deploy models into products and **monitor them in production**
- Work on NLP, computer vision, recommendations and large language models
- Write reliable, production grade code and run experiments

Tools and Technologies

- **Languages.** Python, SQL, sometimes C++ or Java
- **Frameworks.** PyTorch, TensorFlow, scikit learn, Hugging Face
- **Cloud and DevOps.** AWS, GCP, Azure, Docker, Git
- **Data.** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product and software teams

Work environment and personal traits

Hours and setup

- Typically an **office setup**
- Usually **8 hours per day**
- Longer hours may be needed depending on workload and deadlines

Teams and access

- Work alongside **data scientists and IT experts**
- Collaborate to develop applications and systems
- Opportunities exist for people with disabilities

Personality traits

- Enjoy science
- Be creative
- Like using computers
- Have an interest in studying human behaviour

Scholarships and education loans

Scholarships

- Engineering-specific and institute merit scholarships
- Scholarship programmes to check: IndusInd Foundation, Baba Gurbachan Singh, Samsung Star Scholar, NTPC and Foundation for Excellence
- National Scholarship Portal for central, UGC / AICTE and state schemes

Loan options

- Vidya Lakshmi: a portal for viewing, applying for and tracking education loans
- Bank education loans, subject to the lender's eligibility and terms
- Some states, including West Bengal, Odisha and Bihar, offer student-credit schemes

Availability, deadlines and portal arrangements can change; verify current details before applying.

CAREER PROGRESSION PATH

From entry level to leadership

Junior Engineer	Develop and test systems with guidance.
Senior Engineer	Take greater ownership of technical delivery.
Principal Engineer / Engineering Manager	Progress through technical leadership or people management.
Vice President	Lead broader engineering functions and priorities.
Chief Technology Officer	Guide technology strategy across the organisation.

An indicative progression, not a fixed promotion timeline. Roles and responsibilities vary by organisation.

Course fees and expected income

Approximate course fee

₹43,400 – ₹10,00,000

- Indicative course-fee range
- Varies from institute to institute
- Do not assume this is an annual fee; check the full fee structure before enrolling

Approximate monthly income

₹42,000 – ₹1,33,333

- **Per month**; varies by experience, employer and location
- Indicative and subject to change
- Not a guaranteed starting salary or a stage-by-stage pay scale

Figures are indicative, not a current salary survey. Confirm current fees and compare live role-specific salary information.

Employment opportunities

AI engineers work across industry, research and education.

Software and tech companies

Develop AI-enabled applications and systems.

Research facilities

Work in government or private research settings.

Engineering institutes

Contribute to technical projects and research environments.

Institutions to explore

Selected institutions

- **Indian Institute of Science**, Bengaluru, Karnataka
- **University College of Engineering, Osmania University**, Hyderabad, Telangana
- **Indian Institute of Technology Hyderabad**, Telangana

Before choosing a programme

- This list is **indicative only**, not a ranking
- Check the current department, course level and curriculum
- Verify admissions, entrance exams, fees and duration on the institution’s website
- Compare undergraduate and postgraduate options separately

Building employability for the long run

Courses and Certifications

- DeepLearning.AI specialisations by Andrew Ng
- Google, AWS and Azure ML certifications
- Stanford CS231n for vision, CS224n for NLP
- **Kaggle** competitions and Kaggle Learn

Internships and Portfolio

- Research internships at IITs and IISc
- Google Summer of Code and industry internships
- Open source and a **GitHub project portfolio**
- Build things and share them publicly

Future Outlook

- Develop skills in **generative AI and applied ML**
- Keep programming and data fundamentals strong
- Study **responsible and ethical AI**
- Continue learning as tools and roles evolve

Questions & discussion

Thank You